Prateek Komarla

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Education

Rutgers University, New Brunswick NJ

Bachelor of Science in Computer Science and Data Science *May 2027*

Relevant Courses: Data Structures (Java), Linear Algebra, Mobile App Developement, Discrete Structures, Computer Architecture, Data Management for Data Science

GPA: 3.78 out of 4, Deans List

Technical Skills & Certifications

Languages: Java, Python, HTML, CSS, JavaScript, React, Git, MongoDB, SQL
Frameworks/Developer Tools: TensorFlow, Keras, Unity, VSCode, Git, LangChain, Streamlit, Pandas, Docker, Hugging Face
Certifications: Amazon Cloud Practitioner Certification

Experience

Software Engineer Intern May 2025 – August 2025

Astralinx *Remote*

- Developed a business document suggestion platform that provided automated recommendations for grammar, clarity, and professional tone, helping users produce polished business papers efficiently.
- Developed and maintained a Spring Boot backend with an SQL database, implementing RESTful APIs and data processing logic.
- Optimized backend query performance, reducing API response time by 35%.
- Built an interactive TypeScript frontend, enhancing user experience with dynamic interfaces and real-time suggestion updates.
- Integrated natural language processing (NLP) features to generate context-aware editing suggestions.

Rutgers University Mar 2025 – Currently Working

Research Assistant — Image Data Analysis for wound progression in cell biology models *Piscataway, NJ*

- Gathered numerical data from images regarding scratch area over a period of time by examining pixel values and conduct statistical data analysis on Google Sheets/Excel; automated entire image analysis process using python.
- Built a python program to conduct image color thresholding analysis to gather data for biological experiments.
- Designing a convolutional neural network capable of conducting image segmentation to isolate wound regions from surrounding tissue.

Projects

Convolutional Neural Networks for Image Segmentation | *Python, TensorFlow, Keras, Deep Learning, Computer Vision*

- Implemented a symmetrical encoder-decoder U-Net architecture from scratch utilizing the TensorFlow/Keras functional API to achieve end-to-end semantic pixel-level segmentation.
- Formulated a customized pixel-activation classification layer using a 1×1 Convolution with a Sigmoid activation function, establishing a stable continuous probability distribution mapped across more than 1 million individual canvas pixels.
- Created an automated image-padding tracking algorithm to mathematically compute active canvas ratios, normalizing final wound surface area calculations against the structural artifacts of square image transformations.

DataChat | *Python, LangChain, GPT-4o, Streamlit, Pandas, Plotly, ReportLab*

- Built an AI-powered data analyst that lets users upload any CSV or Excel file and query it in plain English, generating and executing pandas code against real data via a LangChain agent.
- Engineered a three-stage chained LLM pipeline that analyzes datasets, writes narrative business reports, and exports them as formatted PDFs using ReportLab.
- Deployed on Hugging Face Spaces via Docker with environment-based secret management

ClimateShield | *Next.js, XGBoost, AWS, Supabase*

- Built a full-stack climate risk intelligence platform that scores wildfire and flood risk (0-100) for any US property address using a custom XGBoost ML model trained on 4 government data sources.
- Deployed ML scoring API on AWS Lambda via Docker/ECR, achieving a wildfire MAE of 5.72 and flood MAE of 7.01 across 3,144 US counties.